

Consistent metropolitan boundaries for the remote sensing of urban land

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ABSTRACT

This paper introduces an internationally consistent definition of metropolitan areas to the literature regarding the remote sensing of urban land use or land cover. In the cross-comparison of land use or land cover for explicitly bounded urban areas, the observed 'economic' definition is argued to hold distinct potential merits over administrative or agglomeration-based boundaries, which typically underpin other studies. To illustrate the proposed merits as well as their implications for the remote sensing literature, the empirical analysis considers the case of 687 European metropolitan areas. Across these metropolitan areas, whose boundaries are defined jointly by the OECD and the European Commission, land cover and land use are segmented in a fusion of imagery from radar and optical sensors in Sentinel satellites. Segmentation is achieved using deep learning in a well-established model architecture. The analytical focus is on built-up areas that are in a residential use or in a commercial or industrial use. Map classifications and accuracy measures are obtained for cities as well as their respective commuting zones as these together embody metropolitan areas. The results underline that not only land use area estimates but also map classification accuracy vary widely across individual metropolitan areas. Whereas classification accuracy to some degree varies for metropolitan areas within as well as between countries, classification accuracy is positively associated with population size and built-up area density as regression analysis confirms. Additionally, the extent of built-up areas in distinct uses is shown to vary across different types of metropolitan (sub-)areas. *This study's findings highlight the typically unobserved role that study area definition and selection may play in affecting outcomes in remote sensing studies in urban settings, as relevant to both studies of single as well as multiple urban areas.* The consistent comparison of remote sensing outcomes across metropolitan areas may further promote generalization in a growing and global field and potentially supports better-informed policy making processes.

1. Introduction

The proliferation of remote sensing as a worldwide source of understanding urban areas majorly draws on technological advances that are distinctively meticulous. Consider advances in sensor platforms (Levin et al., 2020) or the increasingly subtle and advanced workings of classification algorithms (Reba and Seto, 2020). With this, the scientific observation and assessment of urban land cover and land use is deepening and becoming increasingly precise across ever more places (Zhu et al., 2019). However, when comparing outcomes across studies for different places, the important benefits of technical meticulousness are potentially overshadowed by a seemingly elementary matter of study design – that is, the definition and selection of study areas may confound

the comparability of land cover or land use area estimates across distinct urban areas.

Ensuring that comparisons across urban areas are made on a consistent basis requires conceptual consideration of how urban boundaries are delineated. Importantly, the administrative boundaries of many cities and other urban places or regions are substantially and arbitrarily misaligned with the mechanisms that give rise to cohesive urban areas (Moreno-Monroy et al., 2021). Such misalignments obstruct the harmonized comparison and understanding of urban land cover and land use outcomes that are required to appropriately inform academic and policy debates (e.g., OECD, 2022; Seto et al., 2017).

In adopting a more consistent spatial unit of analysis, remote sensing studies of land cover or land use in urban areas –when the extents of

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observed urban areas are explicitly bounded—almost exclusively rely on notions of agglomeration, both in studies of nighttime or daytime imagery (e.g., Mertes et al., 2015; Zhou et al., 2018; Taubenböck et al., 2019; Cao et al., 2020; Li et al., 2020; Zhao et al., 2020; Zhu et al., 2022). Agglomeration-based measures of urban boundaries are usefully specified on the conceptual basis that urban areas are, in essence, concentrations of human activity. These concentrations of human activity arise when centripetal forces outweigh centrifugal forces, as documented in seminal economic literature (Krugman, 1996). However, these forces are not necessarily restricted to areas where built-up features are directly contiguous (Duranton, 2021). This means that cohesive urban areas can extend beyond the (remotely observable) morphology of pure agglomeration.

In this light, the present study adopts a ‘people-based’ view on what bounds an urban area. This type of perspective is well-established in the field of economics, see Duranton (2021), and considers that an urban area as a local economic ‘system’, beyond its physical concentration of built-up structures, is in essence bounded by processes of internal labor market integration (e.g., Bosker et al., 2021; Dijkstra et al., 2021; Moreno-Monroy et al., 2021). An integrated labor market here comprises of an urban core, which consists of contiguous settlements with a high population density and could be referred to as ‘city,’ as well as a surrounding hinterland of lower-density areas whose inhabitants commute to the city. Such functional integration applies to large as well as small-sized urban areas, and so as a mechanism the locational sorting of firms and households as well as the flows of people give rise to what even for smaller places can be considered to be their ‘metropolitan’ boundaries.

The relevance of metropolitan boundaries to the analysis of remotely sensible urban land cover or land use draws in part on the foundational Alonso theoretical model of urban spatial structure (Alonso, 1960). This model crucially observes that households face a trade-off between location rents and the costs of commuting. An implication of these trade-offs, which continue to hold relevance in modern urban economic theory and analysis (Duranton and Puga, 2015), is that the built-up density of an urban area, which results from the response of developers to the intensiveness of demand for space, may fall gradually from a city’s center outwards until land is used only by agriculture, or covered by wilderness. However, as Alonso (1960) notes, there are realistic complications. Commuting zones, besides from their typical coverage across suburbs and small settlements, may also integrate multiple cities or ‘urban cores’ (e.g., Brezzi and Veneri, 2015), rural areas (e.g., Dijkstra et al., 2021), relatively distant satellite towns (e.g., Hui and Lam, 2005), or extend well beyond restrictive land use zoning interventions such as green belts (e.g., Kühn, 2003). Such complications are commonplace as well as potentially beyond the scope of agglomeration-based definitions of urban areas (Duranton, 2021). Crucially, the use of land as well as density are determined interrelatedly between sub-areas due to their functional integration generated by labor market links. These links also overarch sub-areas that may be subject to different policies that affect land use, whether directly or indirectly via the sorting of households and firms into particular locations within the metropolitan area. As such, as a functionally cohesive space, the metropolitan area presents a theoretically desirable unit of analysis when seeking to compare land use across urban areas on a consistent basis.

Relatively few studies on the remote sensing of urban land cover or land use, however, have so far adopted the metropolitan area as their unit of analysis, empirically (e.g., Weber and Puissant, 2003; Yuan et al., 2005; Miller and Shirzaei, 2019; Srivastava et al., 2019; Daams, 2023). One possible reason for this is that the functional integration of non-contiguous sub-areas within a metropolitan area is challenging to observe from information captured by remote sensors. It may even be noted that the literature on defining urban boundaries based on remotely sensed information is thinner than the apparent empirical potential might suggest. In fact, remote sensing studies of urban land, when not resorting to administrative boundaries, commonly do not

explicitly delineate urban areas at all other than, for clear pragmatic reasons, by illustrational square frames, as their study areas cover entire countries or world-regions (e.g., Froliking et al., 2013; Goldblatt et al., 2018; Frantz et al., 2021; Xie et al., 2019). As recently acknowledged in Cao et al. (2020), consistent and broadly accepted metropolitan area definitions remain lacking in the remote sensing literature.¹

This study aims to contribute to the literature by examining the usefulness and possible implications of introducing a consistent measure of metropolitan boundaries in the remote sensing of urban land. To this end, we adopt the definition of metropolitan areas developed jointly by the Organisation for Economic Co-operation and Development (OECD) and the European Commission, see Dijkstra et al. (2019). This EU-OECD definition operationalizes metropolitan areas as high-density urban cores with hinterlands of lower densities that are functionally integrated via commuting flows and is harmonized for international comparative analysis. This offers relevant new ground for consistent cross-metropolitan analysis that promotes consistent comparability, as relevant also to the internationalization, see Reba and Seto (2020), of urban land cover or land use assessments within the field of remote sensing. Relevantly, the present study has three features, integral to its overarching aim, that each allow for a separate additional contribution to the remote sensing literature, as will be motivated after first introducing the study’s empirical structure.

Empirically, our analysis considers two key uses of land, namely residential and commercial or industrial use. These uses are segmented from each other, as well as from other land use and land cover types, in a fusion of EU-ESA Sentinel-1 SAR and Sentinel-2 optical satellite imagery. The observed imagery allows for distinct uses to be associated with granular object-based features of urban land cover. This is achieved using a U-NET deep learning architecture (Ronneberger et al., 2015). This advanced model, which is well-established in the literatures that use machine learning, has seen increasing use in recent remote sensing studies (e.g., Zhang et al., 2021; Martins et al., 2022; Wu et al., 2021). Discussions of the nature and use of deep learning models as compared to other mainstream classification methods are provided in Reba and Seto (2020) and Ye et al. (2018). In focusing on land use rather than land cover, the object-based U-NET model is able to discern that a single land cover type can be present in land patches of different uses, while also mitigating the salt-and-pepper effects that are typically associated with other individual pixel-based classification models (Blaschke, 2010; Belgiu and Csillik, 2018; Zhang et al., 2019). This relatively robust methodological approach to classification permits us to focus on the study’s main and additional contributions. To this end, although the EU-OECD coverage of metropolitan boundaries has recently expanded globally, see Moreno-Monroy et al. (2021) or see Section 5 for further discussion, the present analysis illustratively covers 687 European metropolitan areas across 27 OECD countries based on the availability of reference labels for model training and evaluation. Furthermore, a simplifying factor for the study’s design is that the setting is cross-sectional, rather than change-based as common in other studies, see Reba and Seto (2020), which is sufficient for our analytical aims.

Relevantly, the present study’s three features integral to its overarching aim that each allow for a separate additional contribution to the remote sensing literature are as follows. First, accuracy measures for land use classifications are examined at the policy-relevant spatial scale,

¹ Some remote sensing studies of urban land use or land cover, albeit few in numbers, observe metropolitan boundaries that are sourced from specific national statistical offices. These studies report either on analyses of single urban areas (e.g., Weber and Puissant, 2003; Yuan et al., 2005; Miller and Shirzaei, 2019; Srivastava, Vargas-Muñoz, and Tuia 2019) or analyses of multiple urban areas within a single country (e.g., Chen et al., 2021). Although such country-specific definitions of metropolitan areas are rooted in the labor market rationale that also underpins the present study, differences between their empirical operationalizations remain which limits their comparability across countries.

see OECD (2022), of distinct metropolitan areas as well as by what is perhaps their most relevant feature, the density of built-up areas. The outcomes of that assessment are of potential relevance to the cross interpretation of the cross-urban area consistency of large-scale remote sensing findings or map products (e.g., Brown et al., 2022; Esch et al., 2017; Estoque et al., 2021; Schiavina et al., 2022; Tsendbazar et al., 2021). Moreover, also the findings of remote sensing studies of even only single urban areas may implicitly be influenced by their choice of study area boundaries, should there be an unobserved relationship of classification accuracy with the features –such as density– of the observed area, whether the choice of boundaries is made explicitly or not.

The second additional contribution is the observation of small, medium-sized, as well as large metropolitan areas. Such coverage of differently sized urban areas is called for, among others, in Reba and Seto (2020) and Güneralp et al. (2020) given the tendency of remote sensing studies to cover mostly only the largest of urban areas, leaving relatively smaller places behind in terms of the development of relevant domain knowledge. In comparing classification outcomes across differently sized urban areas, we use an OECD-standardized definition of population size which, together with the measure of metropolitan area boundaries, allows for comparisons by city size on a consistent basis.

Finally, this study's international setting permits the comparison of urban land use as well as variation in related classification accuracy across countries. Although some remote sensing studies in urban settings assess classification accuracy on a country-by-country basis, this does not appear to be a systematic feature of the literature, with evaluations more typically adopting the scale of regions or world-regions (e.g., Wickham et al., 2013; Curtis et al., 2018; Rosier et al., 2022). What is of potential relevance is that, whereas European countries have been covered extensively in studies of secondary land use data, albeit being understudied relative to the US and China in terms of direct remote sensing applications (Reba and Seto, 2020), the cross-country perspective on urban remote sensing outcomes here is of a uniquely consistent nature.

Overall, this study integrates the economic and remote sensing literatures regarding urban areas in a spatially explicit way, through its conceptual and empirical introduction of consistent metropolitan boundaries and several additional analytical features. The analysis as such, considering also research priorities as discussed in Seto et al. (2017) and Zhu et al. (2019), may promote appropriate generalization across urban areas for effective knowledge transfer and better-informed policy making processes.

2. Methods and data

2.1. Internationally consistent definition of metropolitan areas

Across 27 European OECD-member countries, 687 metropolitan areas are observed. Importantly, the definition of metropolitan areas is designed to be consistent across countries. The boundaries of these metropolitan areas are defined using the EU-OECD definition of Functional Urban Areas, whose details are provided by Dijkstra et al. (2019). According to this definition, a Functional Urban Area is composed of two main parts. The first is a 'city' or 'urban core', which consists of a set of contiguous one-km² grid cells of high-population density (at least 1500 people per km²) with a total population of at least 50,000 inhabitants (Fig. 1A). Grid cells and their relative population count are obtained from GeoStat data sourced from Eurostat. The urban core is subsequently adapted to the boundaries of 'local units', the smallest statistical units for which commuting data are available, where most of the population live in the 'city' (Fig. 1B). The second part is the 'commuting zone', which is composed of all other local units where at least 15% of the workforce commutes to the 'city' (Fig. 1C). The Functional Urban Area, or metropolitan area, is the sum of the 'city' and the 'commuting zone' (Fig. 1D). Importantly, a metropolitan area can have multiple 'urban cores', or 'cities', in the case of polycentric metropolitan areas, reflecting the current reality of human settlements.

Due to the delineation's economic foundations, the metropolitan boundaries allow for land cover and land use to be compared on a consistent basis between metropolitan areas – also when these have different population sizes, whether small, medium-sized, large, or very large. All Metropolitan areas, of any size above 50,000 inhabitants in the 27 covered OECD member countries are observed. The study area as such totals a surface of 992,680 km², which encloses a wide variety of urban areas and environments.

2.2. Ground truth reference on urban land use

Urban land use maps that serve as ground truth reference in training the deep learning model and evaluating its classification performance are sourced from the Copernicus Urban Atlas database. These data offer a comprehensive classification of urban land use, as well as further land cover, as measured by granular vectors (Montero et al., 2014). In specific, the vectors observe any patches of land use that are at least 0.25 ha in size as per 2018, which is the latest year of coverage of these maps, across the study area. Details about accuracy and uncertainty are given in Copernicus (2021). Furthermore, land use is defined on an object-basis rather than at the scale of individual pixels. This adopts the

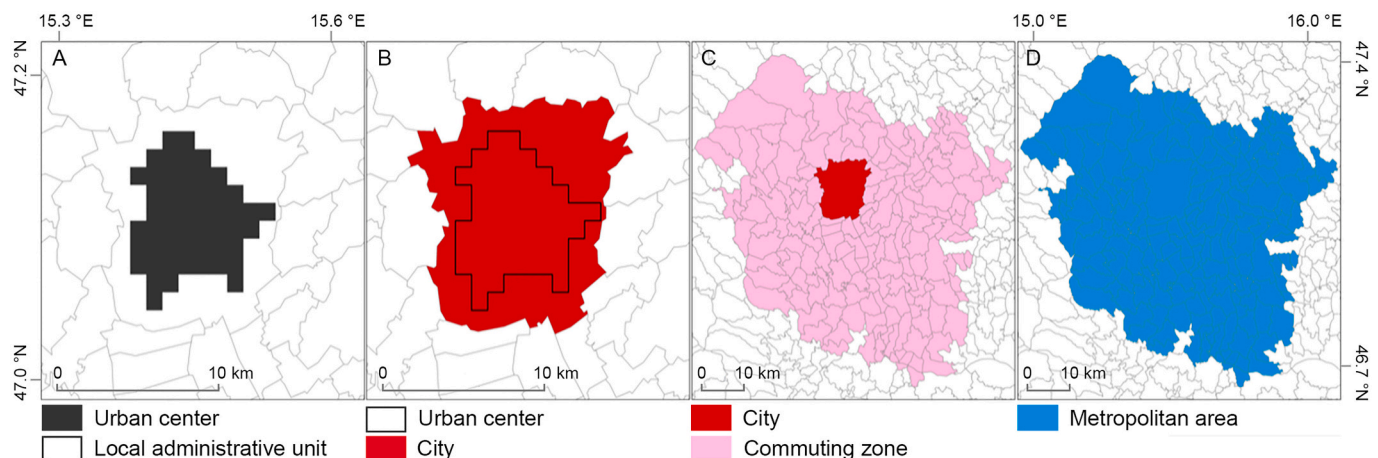


Fig. 1. Consistent delineation of metropolitan areas, as illustrated here for Graz, Austria, which comprise of a city, or cities, and a commuting zone (adapted from Dijkstra et al., 2019).

conceptual view that urban land use in particular locations may consist of patches of land that coherently bundle built-up as well as non-built-up land cover. To allow for a parsimonious and effective segmentation of, in particular, the targeted urban land uses, the original land use and land cover sub-classes in the Urban Atlas maps are aggregated into six classes, which may improve classification (Zhu et al., 2016). When urban residential or commercial uses overlap, the use of the visually dominant and complete object, in both use and shape, is assigned (European Commission, 2020). Class definitions are documented in Table 1, which also provides descriptive statistics for these classes regarding their areal shares at the level of individual metropolitan areas across Europe.

The reference map's resolution is matched with the resolution of the observed satellite imagery (see Section 2.3) in assigning each individual image pixel a label. This overcomes some of the sampling choices that especially remote sensing studies that rely on manually collected data face, see Olofsson et al. (2014) in acquiring labels for model training and evaluation. Furthermore, the use of labels sampled for the pixel-based evaluation of model performance rests on the premise that the quality of the reference classification is superior to the model's classification for the same pixels (Olofsson et al., 2014). In this context, it is noteworthy that Urban Atlas maps are based on imagery of even higher-resolution imagery than analyzed in the present paper and also largely created via manual interpretation, see Montero et al. (2014), meaning that the assumption that the observed reference classification is sufficient for this study's purposes is reasonable.

2.3. Fusion of Sentinel-1 and Sentinel-2 satellite imagery

The imagery observed in this study come from the Sentinel-1 and Sentinel-2 satellite constellations operated by the European Commission and the European Space Agency. Following other studies of urban land use or land cover the radar-based and multi-spectral imagery from these satellites is fused to increase the variation in information that can be observed in the classification model (Li et al., 2020; Zhang et al., 2021; Zhu et al., 2022). In specific, selected Sentinel-1 and Sentinel-2 image bands are fused by stacking these bands such that the tensors that serve as input data for the deep learning model are composed of 8 bands, as discussed in more detail next.

Sentinel-2's multi-spectral optical imagery is this study's base source of remotely sensed information. In specific, Level-2A pre-processed imagery with a base resolution of 10×10 m is obtained for 2018 and filtered for clouds and cloud-shadows using Google Earth Engine

Table 1
Operationalization of the targeted land use and land cover classes.

Target Class	Original Urban Atlas Sub-Classes	Mean (%)	St. dev. (%)
Residential built-up area	Discontinuous or continuous urban fabric, isolated structures.	7.8	5.3
Industrial or commercial built-up area	Industrial, commercial, public, military, or private units, mineral extraction and dump sites, construction sites, land without current use.	4.1	3.3
Transport infrastructure	Fast transit roads, other roads, railways, port, airports.	2.7	1.6
Open space	Forests, herbaceous areas, non-vegetated open space (beaches, bare land), green space, sports or leisure facilities.	33.8	19.3
Agricultural	Arable land, permanent crops, pastures, complex or mixed cultivation, orchards.	48.8	19.1
Water or wetlands	Water, wetlands.	2.7	4.3

Notes: mean and standard deviation values are for the shares observed across individual metropolitan areas.

(Gorelick et al., 2017). Only images with cloud cover below 60% are used, and their average cloud cover is 19.5% prior to the removal of clouds and cloud shadow areas from each individual image. Subsequently, to further reduce the possible influence of any remaining clouds, haze, or shadows, pixel-level median values, across all images at the year-level, are taken. These median values are obtained for each Sentinel-2 band, as well as for each of the Sentinel-1 bands that will be described shortly, separately. For Sentinel-2 images, median values are obtained across images from the Sentinel-2A and Sentinel-2B satellites conjunctively, as done similarly conjunctively also for Sentinel-1A and Sentinel-1B imagery. Imagery acquisitions cover the entire observed year, although for polar regions only the May to October imagery is taken into account to reduce the observed presence of snow. The selection of spectral bands includes RGB, near-infra red (NIR), and short-wave infrared (SWIR1 and SWIR2) bands. NIR values may be particularly useful in capturing vegetation, to segment such land cover from built-up urban land uses where relevant (Huo et al., 2021). SWIR values offer additional information, as useful for instance in separating water bodies or vegetated open space from more typical forms of urban land cover. Some studies add additional indices, such as vegetation indices, that are generated based on cross-band calculations (e.g., Goldblatt et al., 2018). This practice is not adopted here as cross-band relationships are assumed to be picked up by the deep learning model (see Section 2.2) and in part for parsimony.

The optical spectral bands are, however, complemented with bands from Sentinel-1C-band synthetic aperture radar imagery. These images are sourced from Level-1 Ground Range Detected scenes, that are pre-processed in Google Earth Engine using the Sentinel-1 Toolbox, which involves several steps. These steps, see also Hafner et al. (2022), obtains each pixel's backscatter coefficient by accounting for ESA information on precise orbital paths, low intensity noise at the edges of scenes, thermal noise that could otherwise lead to discontinuities between sub-swaths, besides applying radiometric calibration and orthorectifying the imagery using the STRM 30 m DEM or, for metropolitan areas at latitudes upwards of 60° , the ASTER DEM. Backscatter intensity is converted into decibel values via log-scaling ($10\log_{10}x$). From the resultant imagery, VV (i.e., vertical transmit and vertical receive) and VH (vertical transmit and horizontal receive) bands are selected. These bands add information, see Li et al. (2020) or Zhu et al. (2022), regarding in particular the height and structure of land cover and are insensitive to cloud cover or other weather circumstances due to their radar-based measurement. The complementary usefulness of these data is also apparent from the acquisition times of the individual images, of which 72.5% and 58.8% are during the official March–October European summer period for Sentinel-1 and the relatively more weather-sensitive Sentinel-2 imagery, respectively. In total, 151,343 Sentinel-2 images and 222,317 Sentinel-1 images are observed. After integration of the Sentinel-1 and Sentinel-2 imagery, the imagery is sliced into smaller images ($N = 622,313$) of $160 \times 160 \times 8$ dimensions.

2.4. Deep learning model for map classification

Classification is performed using a deep learning model that follows the well-established U-NET architecture (Ronneberger et al., 2015). This model architecture uses convolutional neural networks, similar to models in other recent studies of remote sensing in an urban context, see Martins et al. (2022), Zhu et al. (2022), Sirko et al. (2021), or Srivastava et al. (2019), to segment patches of land use. This is consistent with the conceptual premise that patches of land use may internally consist of single or multiple types of land cover, depending on their local relative situatedness as well as their 'object-based' coherence.

In detail, as visualized in Fig. 2, the model architecture follows a symmetrical U-shape of convolutional neural network layers. To capture context, a classical contractional path uses 3×3 convolutions followed by a rectified linear unit function and a 2×2 max pooling operation with strides of 2 for down-sampling. At each step of down-sampling, the

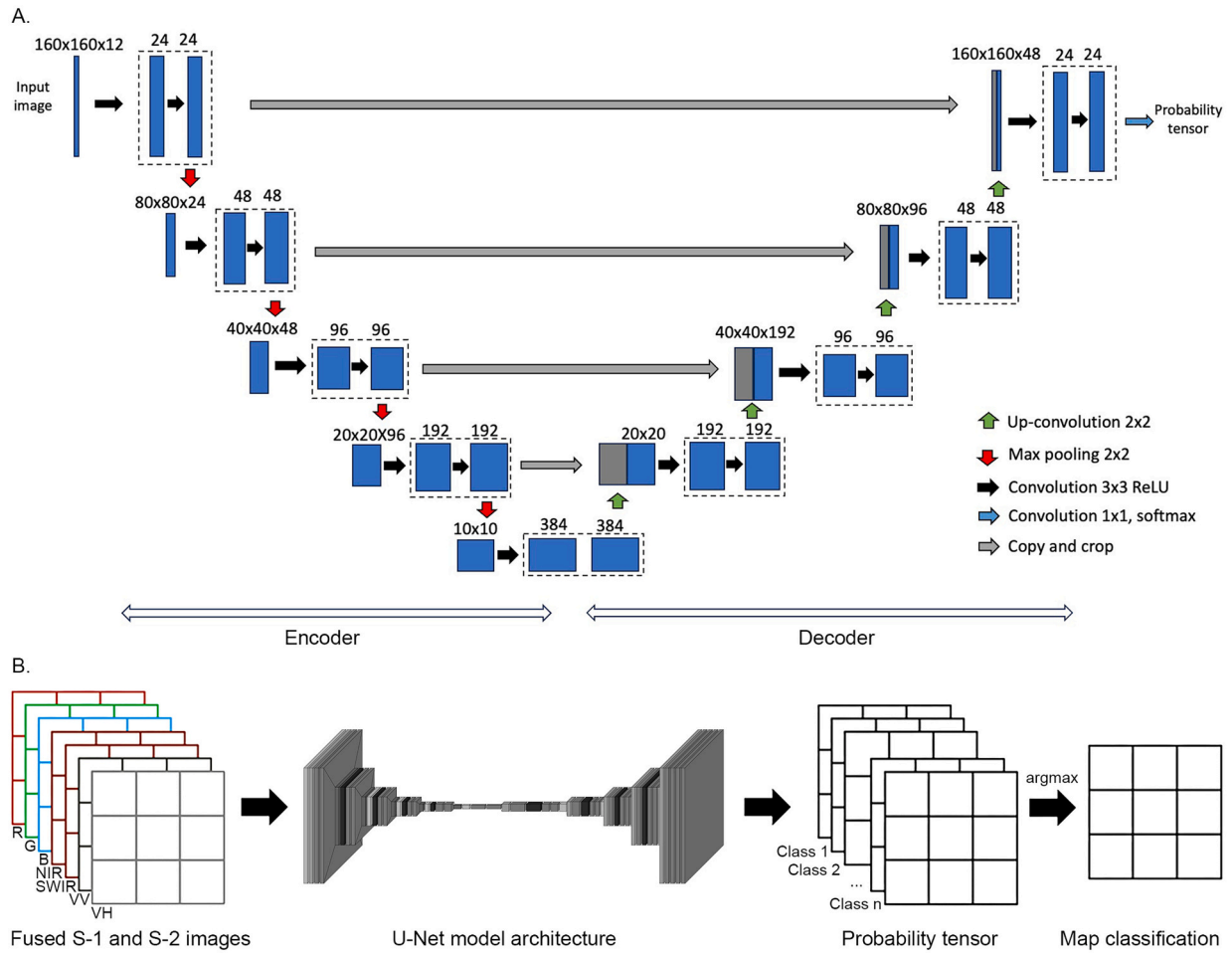


Fig. 2. Schematic visualization of the deep learning-based approach. Including (A) the U-NET model's key parameters and (B) an overview of the classification procedure.

number of feature channels is doubled. In the subsequent path of expansion, which may capture the localization of patterns, an up-sampling of the feature map is followed by a 2×2 up-convolution halving the number of feature channels, a concatenation with the correspondingly cropped feature map from the contracting path, and two 3×3 convolutions that are each followed by a rectified linear unit function. In total, the network has 23 convolutional layers. After each layer, a dropout of 0.25 is applied. At the final layer, a 1×1 convolution is used to assign each 64-component feature vector to the observed number of land cover and land use classes.

The model architecture as well as further steps that are part of the satellite imagery classification procedure are illustrated in Fig. 2 as well. As reflected in Fig. 2, the set of image bands is constant while classifying each pixel. The classification's initial output is a probability tensor. This tensor consists of a three-dimensional array that for each pixel gives the probability that it belongs to a particular class. Final class assignment is obtained by assigning an observed pixel to the class with the highest probability as observed among the tensor's class-specific probability layers.

The configuration of the deep learning model in terms of training and validation is as follows. The deep learning model is trained in a supervised manner. To this end, the observed set of images is split into subsets for training as well as for validation and testing separately using a 60–20–20 split allocation. The split allocation is random, which is reasonable as the coverage of the large sample covers the entire study area which mitigates possible unobserved selection effects from sampling land use patches of different shapes and sizes, which is an ongoing

subject of debate in object-based classification (Ye et al., 2018; Stehman and Foody, 2019). Furthermore, data augmentation consisting of horizontal and vertical random flips, random rotations of 90° , and random contrast is applied during model training. The test-set is used at the end of each training epoch to assess the model's intermediate performance and to avoid overfitting. An early-stopping criterion tracks at the end of each epoch if the loss, as measured using a sparse cross-entropy loss function, reaches a maximum. It may be noted that the set referred to as the 'validation set' should not be confused with the set that is ultimately used for the assessment of the final classification's accuracy in the main analysis, which in line with the deep learning literature is referred to here as the 'test set.' This test set is independent from the sets that are observed during the model's learning process, in line with common standards in obtaining reference samples (Olofsson et al., 2014).

2.5. Map accuracy evaluation metrics

To evaluate the accuracy of metropolitan map classifications a confusion matrix is used as well as several standard metrics, namely precision, recall and the F1-score. These metrics are obtained for each c th land use class of interest separately for the observed metropolitan area m ($m = 1, \dots, M$) as follows. Precision, also commonly referred to as user's accuracy, can be specified as:

$$p_{c,m} = \frac{TP_{c,m}}{TP_{c,m} + FP_{c,m}}, \quad (1)$$

where p denotes precision, TP denotes the count of true positive

classifications and FP the count of false positive classifications, with all classifications being observed among pixels in the test set. Similarly, recall, or producer's accuracy, can be written as:

$$r_{c,m} = \frac{TP_{c,m}}{TP_{c,m} + FN_{c,m}}, \quad (2)$$

where r denotes recall and FN is the count of false negatives. The F1-score, in turn, can be written as:

$$F1_{c,m} = 2 \frac{p_{c,m} r_{c,m}}{p_{c,m} + r_{c,m}}, \quad (3)$$

where $F1_{c,m}$ denotes the harmonic mean of precision and recall values. Furthermore, to account for variation in classification accuracy at the level of individual metropolitan areas, following recommendations in Olofsson et al. (2014) and Stehman and Foody (2019), different from 'counting pixels,' mapping errors are adjusted for in estimating the areas of the land use classes of interest.

3. Results

3.1. Metropolitan area-scale assessment of mapping accuracy

Before turning to the main cross-metropolitan analyses that are at the heart of this study, the accuracy of the deep learning model's map classification can be examined from Table 2. Table 2 shows a confusion matrix based on the counts of reference pixels as cross-tabulated by predicted class and the actual class according to reference data. This information indicates that, for instance, residential built-up area is classified with a producer's accuracy, or recall, of 0.84 and a user's accuracy, or precision, of 0.72, whereas these accuracy values are 0.90 and 0.93 for agricultural land and 0.92 and 0.88 for open space, respectively. For transport infrastructure, which in effect is merely a 'control' class besides the built-up land uses of interest, particularly low accuracy values are observed. As initial examinations of the map classification alongside a mapping of assignment probabilities for the transport class confirmed, infrastructure networks are classified granularly, but also blend into other closely surrounding urban uses for which assignment probabilities are higher. Therefore, as pixels are assigned to the class for which the highest probability value is observed across all bands, land cover associated with transport is typically overshadowed by built-up classes, despite the relatively high resolution of the satellite imagery.

Overall, the observed classification accuracy levels are in range with accuracies reported in other studies of distinct urban land use or land cover classes as well as with accuracies in studies that observe a single and thus relatively straightforward to classify built-up area class (Du et al., 2021; Li and Gong, 2016; Goldblatt et al., 2018). Although accuracy comparison is limited due to cross-study variation in both class definitions and degrees of accuracy-reporting as Reba and Seto (2020) observe, the similar accuracies do indicate the deep learning model's appropriate ability to distinguish built-up areas as differentiated by the urban land uses of interest, namely residential use and industrial or

commercial use, for the present study's purposes. Visual illustrations for the cities of Amsterdam, Vienna, and Berlin are provided in Fig. 3.

Regarding the land uses of interest, it should furthermore be noted that, in line with findings in Du et al. (2021), study area-wide precision and recall values for residential built-up area (0.72 and 0.84, respectively) are both noticeably higher than those for industrial or commercial built-up area (0.67 and 0.66 respectively), as Table 2 reports. This observation will importantly be further explored next, at the level of individual metropolitan areas.

At the level of individual metropolitan areas, map classification accuracy varies considerably. Fig. 4 plots precision values (user's accuracy) against recall values (producer's accuracy) for residential built-up area (Panel A) and industrial or commercial built-up area (Panel B). In both panels, most of the metropolitan areas are positioned in the panel's top-right quadrant, which

reflects that the map classifications are relatively accurate. It may be noted that classification accuracy varies between residential and industrial or commercial uses in terms of both levels and relative dispersion. The mean of recall values observed for individual metropolitan areas is 0.83 for residential use and 0.64 for industrial or commercial use with standard deviations of 0.08 and 0.11, respectively, whereas mean precision values are 0.70 for residential use and 0.66 for industrial or commercial use with standard deviations of 0.07 and 0.09, respectively. This suggests that, within metropolitan built-up areas, commercial or industrial land use structure is more heterogeneous and more challenging to capture than residential land use structure.

Relevantly, from Fig. 4, precision-recall profiles of individual metropolitan areas can be examined. For both of the observed uses, these precision-recall profiles roughly follow a 45-degree angle. This means that, for an observed use, precision and recall values are typically largely similar. However, across metropolitan areas precision-recall profiles show substantial variation, with some metropolitan areas being associated with higher recall than precision, or vice versa. This importantly points to cross-metropolitan variation in classification accuracy that is of relevance to the interpretation of urban land use mapping outcomes.

For instance, the results underlying Fig. 4 indicate that in Valencia (Spain) residential built-up areas are classified with a precision of 0.68 and a recall of 0.84, with these values being 0.74 and 0.91, respectively, in Strasbourg (France), 0.82 and 0.92 in Krefeld (Germany), and 0.71 and 0.67 in Tromsø (Norway) or 0.85 and 0.95 in Trenčín (Slovakia). Furthermore, relative outliers in terms of metropolitan areas-specific precision-recall profiles can for instance be observed for the class of industrial or commercial built-up areas, with precision and recall values as low as 0.42 and 0.17, respectively, for Saint Denis (France), 0.31 and 0.72 in Greater Soest (the Netherlands), or as high as, for instance, 0.86 and 0.86 in Celle (Germany) or 0.88 and 0.93 in Battipaglia (Italy), asides from the relatively more common-range yet varied accuracy values observed across the metropolitan areas. Whereas the illustrations so far arbitrarily refer to metropolitan areas from each of the distinct OECD-standardized population size categories, the role of metropolitan size will be examined more explicitly shortly.

Another natural consideration to examine first is whether the

Table 2
Confusion matrix for the classification results ($N = 2.6$ billion reference pixels).

		Reference class						
		Residential	Industrial or commercial	Transport infrastructure	Open space	Agricultural	Water or wetlands	User's acc. (precision)
Predicted class	Residential	159,661,202	14,652,659	19,536,902	10,062,739	16,938,242	202,243	0.72
	Industrial or commercial	7,368,453	60,568,266	6,881,463	7,153,064	7,782,185	423,423	0.67
	Transport	1,517,703	2,186,217	18,383,369	2,558,469	4,767,641	203,604	0.62
	Open space	7,198,618	7,016,107	7,607,580	875,981,993	85,342,209	7,255,574	0.88
	Agricultural	13,349,176	7,488,574	13,830,028	49,533,950	1,110,008,208	4,469,085	0.93
	Water or wetlands	147,779	315,129	168,476	2,667,761	1,929,691	72,970,218	0.93
	Producer's acc. (recall)	0.84	0.66	0.28	0.92	0.90	0.85	0.88

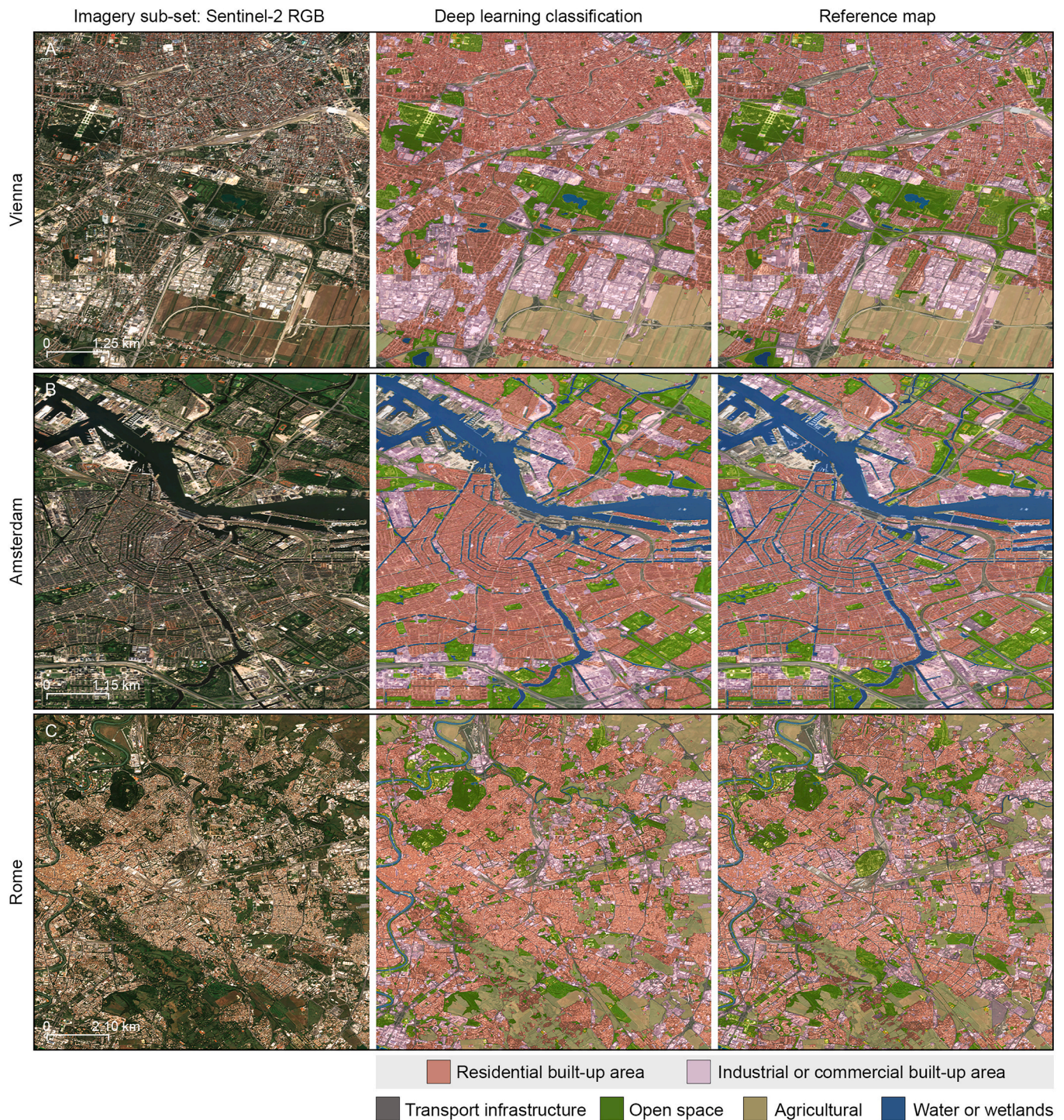


Fig. 3. Visualization of classification results for selected metropolitan areas; Vienna, Austria (Panel A), Amsterdam, the Netherlands (Panel B), and Rome, Italy (Panel C).

physical shape of metropolitan areas may influence how accurately land uses are classified. To ensure purity of approximation, shape here is obtained from the ratio of the built-up area within the total area of the city, or cities, within an observed metropolitan area. This measure then captures urban form in those areas where such form is potentially the most heterogeneous as well as where urban uses of land are relatively concentrated in space. The a-priori relationship between built-up area density and land use classification accuracy is ambiguous. For instance, a relatively high level of built-up area density may make city structure more complex, as land uses may be spatially mixed. At the same time,

high levels of built-up area density may also reflect zoning policies in which distinct uses of land are tightly separated and so could potentially be segmented in empirical classification more effectively. A relatively low level of built-up area density, on the other hand, may imply that the spatial cohesion of different types of land cover becomes more important to signaling a particular use. Compare, for instance, a suburb with widespread vegetation, perhaps including heterogeneous and vast estates or industrial zones, or leafy trees that might obscure built-up structures, to a high-density residential area that is homogeneous in terms of land cover. Nevertheless, empirically, the relationship between

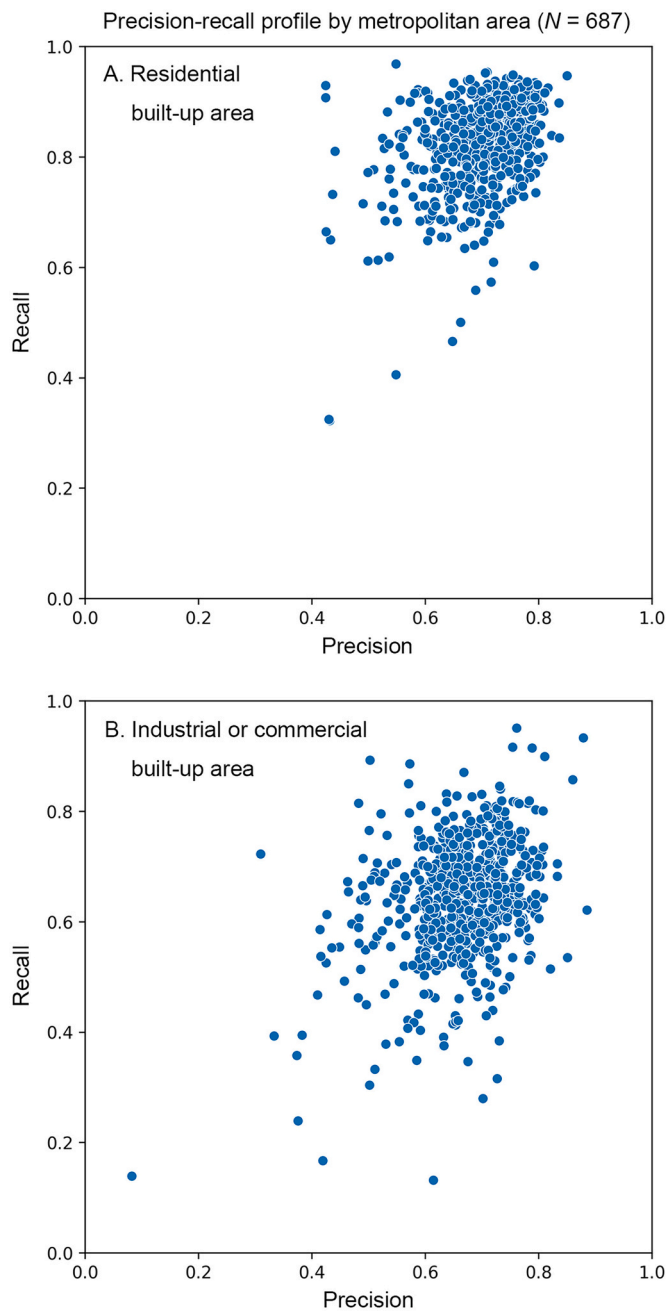


Fig. 4. Precision-recall profiles for each metropolitan area and by class of interest.

city shape and classification accuracy outcomes appears to have clear direction. Fig. 5 plots the measure of city shape, based on total built-up area density, against F1-scores, which represent the harmonized mean values of precision and recall values for the observed city. In both Panels A and B, for residential and industrial or commercial use, respectively, the relationship between built-up area density and F1-scores is positive. This observation suggests that the observed map classifications of urban land uses are typically more accurate in metropolitan areas that have higher built-up area densities.

Furthermore, in Fig. 5, Panels A and B, the observed positive relationship between F1-scores and built-up area density is steeper for residential use than for industrial or commercial use. This potentially relates to the extent to which these distinct land uses are allocated centrally within cities. This may influence use-specific variation in built-up area densities across metropolitan areas. In this context, it can be

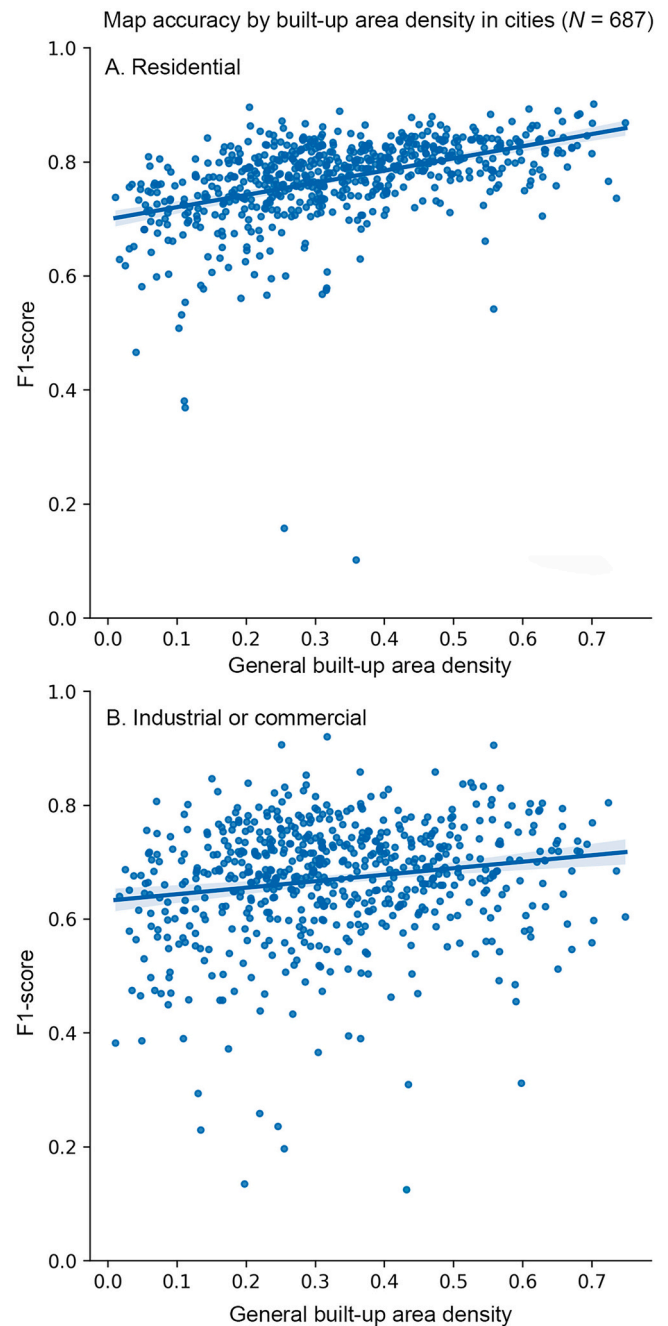


Fig. 5. Map accuracy (F1-scores) varies with urban core built-up area density.

observed from Tables A.1 and A.2 in the Appendix that, for both land uses of interest, map classification accuracy is higher in cities than in commuting zones, whereas non-built-up classes are better segmented in commuting zones. These observations highlight that the shape of cities may influence mapping accuracy.

In addition to comparing accuracy by individual metropolitan area, the underlying accuracy outcomes can also be aggregated for evaluation at the country-level (Fig. 6). In specific, the bars in Fig. 6 show country-specific F1-scores and error bars indicate a 95% confidence interval as related to within-country variation in metropolitan F1-scores. Across countries, F1-scores vary from 0.67 to 0.81 for residential built-up areas and from 0.47 to 0.70 for industrial or commercial built-up areas. Although there are exceptions to a mostly modest range of error bars, such as for Norway, Latvia, Estonia, and Lithuania as well as Portugal and Austria, which does not appear to be driven to any substantial extent

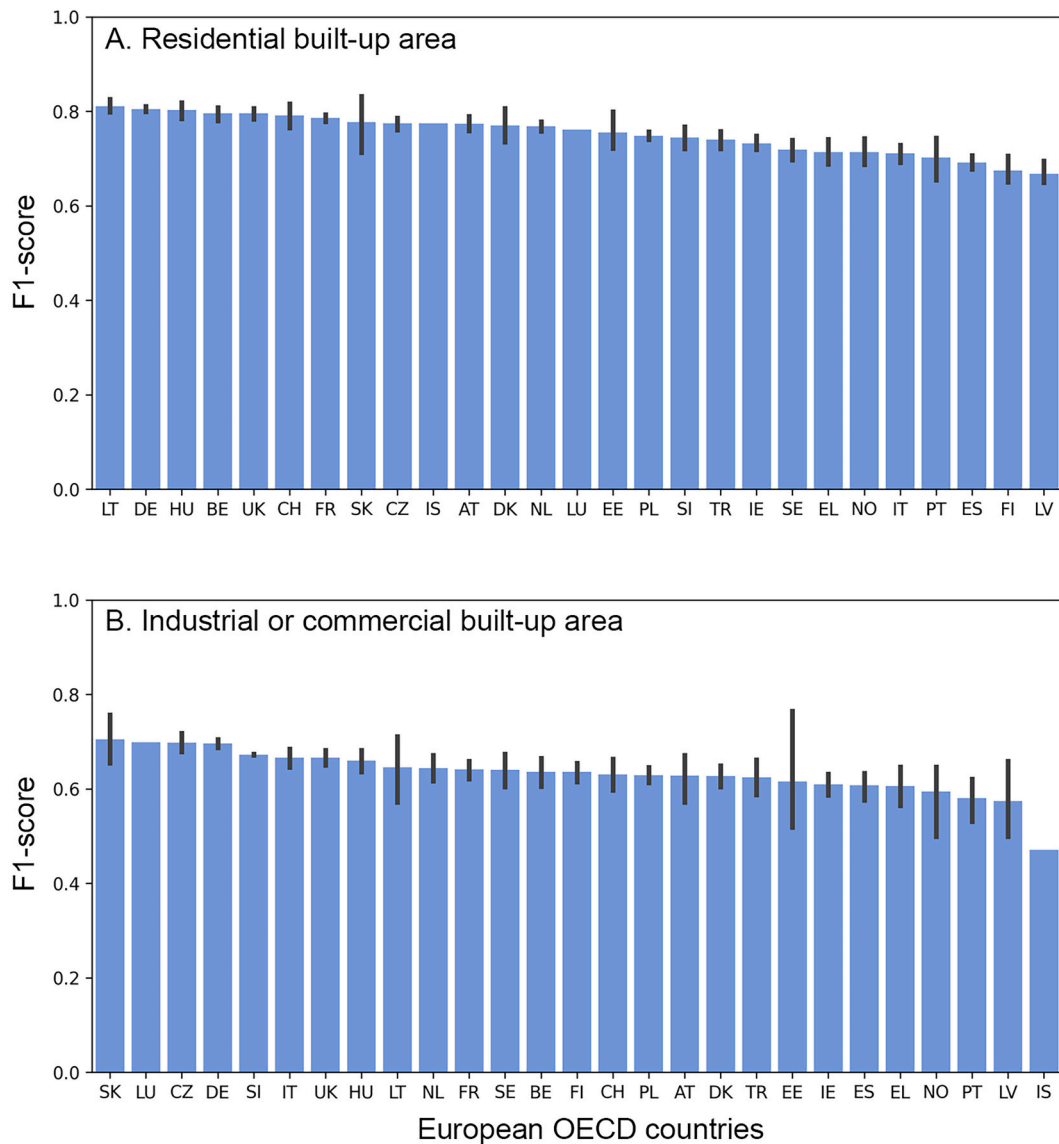


Fig. 6. Average F1-scores for metropolitan areas grouped by country highlight between-country and within-country variation in built-up land use mapping accuracy.

by climatic circumstances (see Fig. A.2 in the appendix), classification accuracy is mostly relatively constant across metropolitan areas within single countries.

The observations here can be further established in a model that

regresses classification accuracy on the features of metropolitan areas. The model's full specification for metropolitan area i ($i = 1, \dots, n$) in country c ($c = 1, \dots, C$), as estimates by use separately, can be written as:

Table 3

Classification accuracy regressed on the features of metropolitan areas.

	Residential built-up area accuracy				Industrial or commercial built-up area accuracy		
	(1)	(2)	(3)		(4)	(5)	(6)
Medium-sized	0.024*** (0.008)	0.032*** (0.008)	0.020*** (0.007)		0.003 (0.011)	0.010 (0.011)	0.006 (0.011)
Large-sized	0.051*** (0.008)	0.054*** (0.008)	0.024*** (0.007)		0.019* (0.011)	0.022** (0.011)	0.013 (0.012)
Very large-sized	0.075*** (0.012)	0.066*** (0.012)	0.042*** (0.010)		0.043*** (0.017)	0.035** (0.016)	0.027 (0.017)
City density	–	0.195*** (0.024)	0.186*** (0.023)		–	0.166*** (0.034)	0.183*** (0.039)
Constant	0.718*** (0.007)	0.683*** (0.008)	0.727*** (0.021)		0.636*** (0.010)	0.605*** (0.012)	0.594*** (0.034)
Country FE	No	No	Yes		No	No	Yes
Adjusted R ²	0.091	0.177	0.477		0.014	0.050	0.145

Notes: In each of the specifications for distinct models across columns (1) to (6), the dependent variable is classification accuracy (F1-scores) at the level of individual metropolitan areas ($N = 687$). The reference category for metropolitan size is small-sized metropolitan areas. The OECD definition of metropolitan size distinguishes between small-sized (50,000 to 100,000 inhabitants), medium-sized (100,000 to 250,000 inhabitants), large-sized (250,000 to 1500,000 inhabitants), and very large-sized (1500,000 or more inhabitants) metropolitan areas. City density is measured consistently with the built-up area density metric in Fig. 5. Country-level clustered standard errors are in parentheses. Results are robust to using beta regressions, see Ferrari and Cribari-Neto (2004), with a logit link. *, **, *** Significance at 10%, 5%, and 1%, respectively.

$$Accuracy_{i,c} = \alpha + \beta_1 City\ density_{i,c} + \sum_{a=1}^A \beta_a Metropolitan\ size_{i,c} + \mu_c + \varepsilon_{i,c}, \quad (4)$$

where α is the constant; $Accuracy_{i,c}$ is the F1-score; $City\ density_{i,c}$ captures the ratio of built-up area in total city surface; and $Metropolitan\ size_{i,c}$ a vector of A OECD-standardized categories of metropolitan area size in terms of population; μ_c captures spatial fixed effects through country dummies; and $\varepsilon_{i,c}$ denotes country-level clustered standard errors. Regression results are presented in Table 3 by type of built-up area, in columns (1) to (3) and columns (4) to (6), in an incremental fashion with increasingly restrictive specifications.

Table 3, columns (1) to (6), shows that classification accuracy responds to metropolitan size. In specific, for incrementally larger metropolitan areas in terms of population, on average, relatively higher classification accuracies are observed, and largely persistently so across built-up area types. In the case of residential built-up areas, compared to accuracy-levels in the reference category of small-sized metropolitan areas F1-scores, as measured on a 0 to 1 interval, are approximately 0.04 to 0.07 points higher across specifications, as can be interpreted in relatively terms here based on a one-unit change in the underlying dummy variable. A similar albeit weaker positive association between metropolitan size and accuracy is observed for industrial or commercial built-up area classifications. Moreover, this association, in contrast to the association observed for residential built-up areas, ultimately is absorbed by controls for fixed effects at the country level. This suggests that for industrial or commercial built-up areas the metropolitan size issue, more so than for residential built-up areas, is relatively bound between-country rather than within-country variation in metropolitan area composition. Nevertheless, the positive relationship between city density, as captured from any built-up area across densely populated urban cores or cities within metropolitan areas, in line with Fig. 5, holds across both of the observed land use types, also when controlling for country-level fixed effects (columns 3 and 5). Furthermore, as adjusted R^2 levels suggest, country effects overall are relevant in explaining variation in map classification accuracy, in particular for residential built-up area.

In the next sub-section, variation in mapping accuracy across individual metropolitan areas is exploited in an examination of area estimates for metropolitan land use. This examination serves to further highlight the usefulness of the consistent definition of metropolitan boundaries as well as potential implications for the remote sensing literature. This illustrational analysis considers area estimates based on classifications in 2021 imagery as obtained per Section 2.3's procedures and under the assumption that metropolitan area-specific rates of accuracy are constant over time. The analysis from here onwards will, other than the previous accuracy-related observations, consider the possible influence of study area definition on the land use classifications themselves.

3.2. Internationally consistent comparison of land use

As relevant to study area selection and definition, this sub-section illustratively highlights several key characteristics of remotely sensed urban land use area estimates at the metropolitan level, as structured along a discussion of observations A to C. Important for this exercise is that the observed definition of metropolitan boundaries allows land use outcomes to be compared consistently across metropolitan areas, also when these are located in different countries, as well as by population size and internal structure in terms of cities and commuting zones.

For an initial overview of the findings discussed next, for each of the observed metropolitan areas, the estimated quantity of built-up area in either a residential use or an industrial or commercial use is shown in Fig. 7. The estimates are expressed on a per capita basis, given the number of inhabitants in the year of classification, following Dijkstra

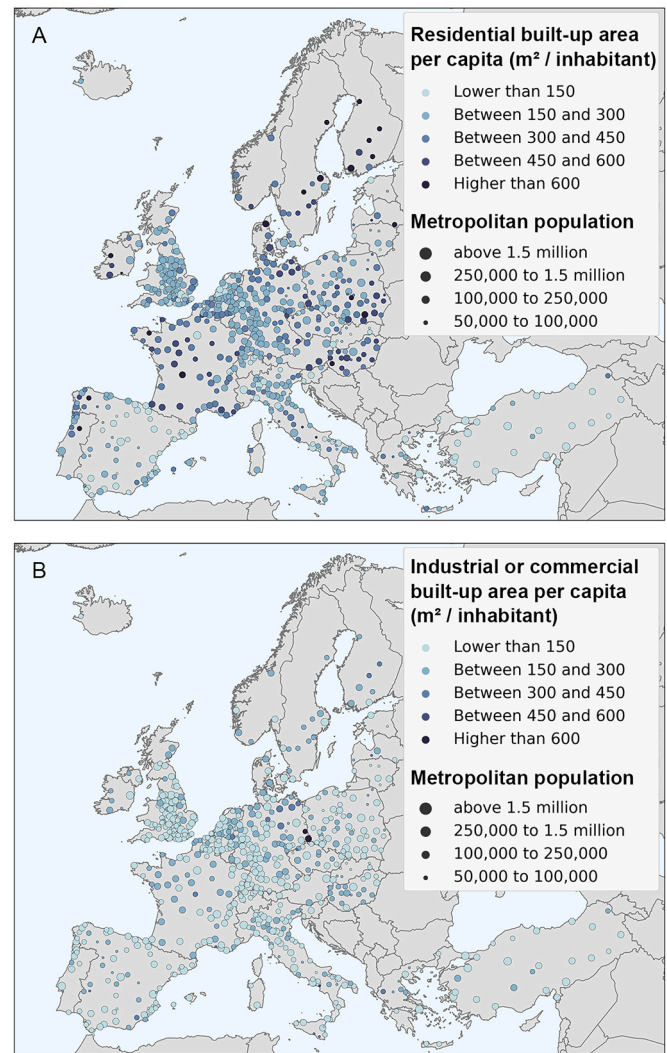


Fig. 7. Estimated metropolitan land use areas per capita.

et al. (2019). This mitigates some remaining influence that areal extents of small local administrative units used in defining metropolitan boundaries otherwise might have, in case built-up areas were expressed in terms of total area alone. Per capita estimates furthermore account for variation in density and so offer a relatively consistent unit of analysis in comparing quantities of built-up area across metropolitan areas. As Panels A and B in Fig. 7 evince, in Europe there is noteworthy between-country as well as considerable within-country variation in residential and industrial or commercial built-up area per inhabitant, as will be explored in more explicit terms next.

A. Residential and industrial or commercial built-up area per capita vary widely between metropolitan areas and countries but can be bounded and compared on a consistent basis—Fig. 8 shows for each of the 27 observed countries the average metropolitan built-up area per capita, as well as within-country variation across metropolitan areas, by type of land use. As already suggested by the maps in Fig. 7, between countries the range of built-up area per capita is substantial. For built-up areas in a residential use (Fig. 8, Panel A), the lowest country-level.

Average of built-up area per metropolitan inhabitant is observed for Turkey at approximately 120 square meters, whereas the highest per capita area is close to 600 square meters per inhabitant, as observed for Finland. Between these outer values, Fig. 8 also shows that the distributional slope when the estimates are rank-ordered across countries is relatively gradual. This indicates that, while there is substantial variation in terms of residential built-up area per inhabitant within Europe,

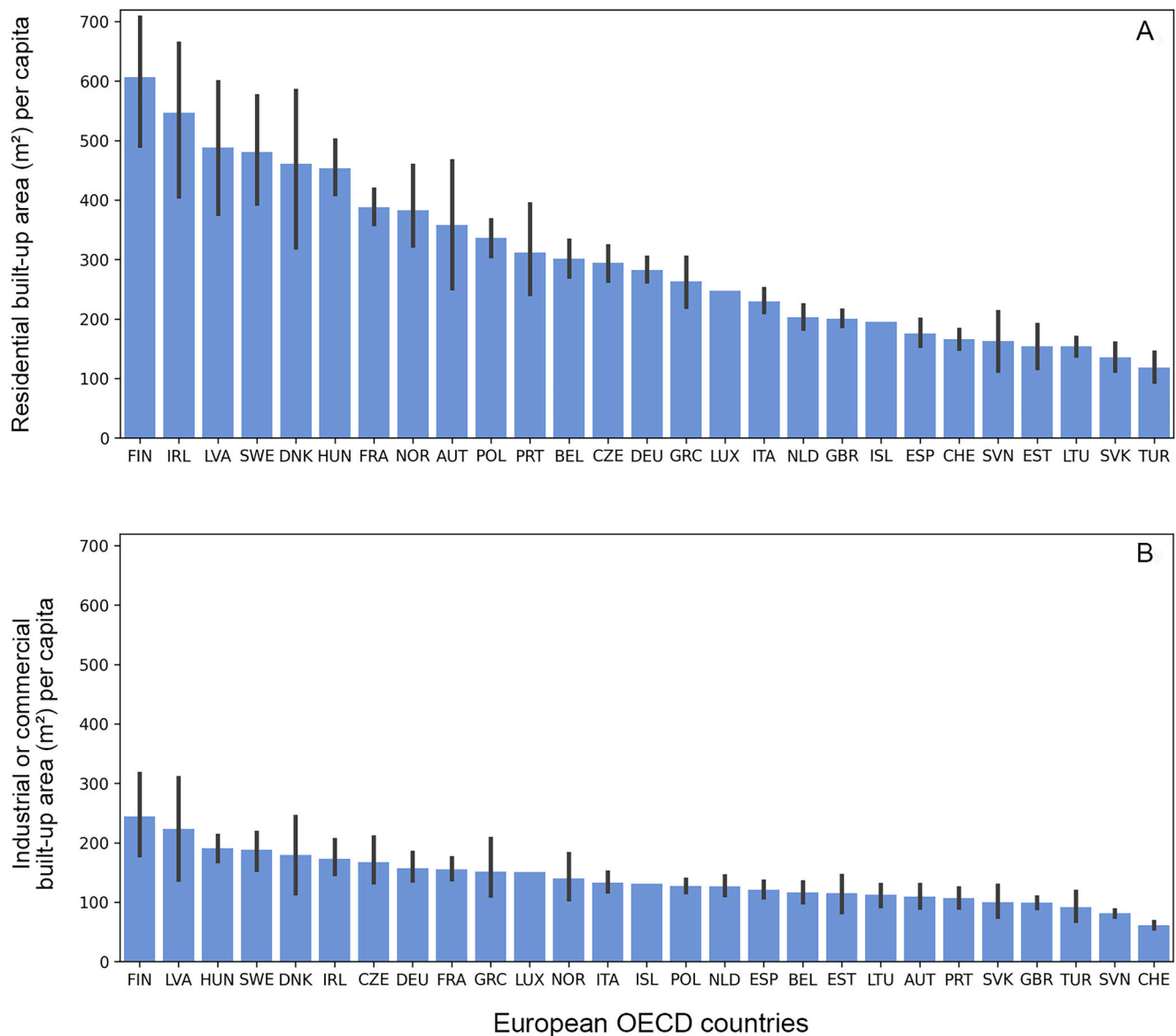


Fig. 8. Built-up area in a residential use (Panel A) and built-up area in an industrial or commercial use (Panel B) on a per capita basis and by European OECD country.

albeit with some regional differences (cf., Fig. 7), there is no clear general divide between countries.

On a country-by-country basis, however, it can be observed from Fig. 8, Panel A, that the residential built-up area per capita in Ireland's metropolitan areas is, on average, about twice that of metropolitan areas in Germany, or smaller for metropolitan areas in Spain and Estonia than for metropolitan areas in countries that are widely known for their compact urban development, such as the Netherlands and the United Kingdom. Largely similar patterns are observed, in Fig. 8, Panel B, for industrial or commercial built-up area per capita, also in terms of how countries rank. Furthermore, several countries show relatively wide variation in built-up area per capita for both industrial or commercial and residential uses, pointing to regional variation in planning cultures or land market structures that are consistent across the different uses.

B. A study area's spatial definition matters: within urban areas, commuting zones rather than cities may drive estimates of built-up area per capita—In Fig. 9, the observed estimates of built-up area per capita are now further disaggregated, namely by internal metropolitan structure as comprised of cities and commuting zones. This is necessary to appropriately interpret built-up area per capita in the light of economic

explanations of how intensively or extensively land is used, in the vein of extensions of the Alonso model of urban land markets. In particular, Fig. 9, Panels A and B, show how the metropolitan area-level built-up area per capita estimates considered so far are upwardly driven by relatively high levels of built-up area per capita in commuting zone. An economic explanation for this is that land values in commuting zones tend to be relatively low.

As compared to land values in cities, where there is more competition for space. As such, land in commuting zones tends to be used relatively extensively rather than relatively intensively. This underlines the usefulness of the metropolitan boundary definition used in the present study in comparing remotely sensed built-up area information, in quantities, across different urban locales.

Furthermore, turning back to the empirical evidence in Fig. 9, several of the countries for which in Fig. 8 some of the largest built-up areas per capita were observed across their overall metropolitan area surface (e.g., Ireland and Latvia) are now shown to actually include cities with relatively low built-up area per capita. This suggests that in those countries in particular, the metropolitan area-wide outcomes are primarily driven by spread-out land use in commuting zones. Noteworthy is that large

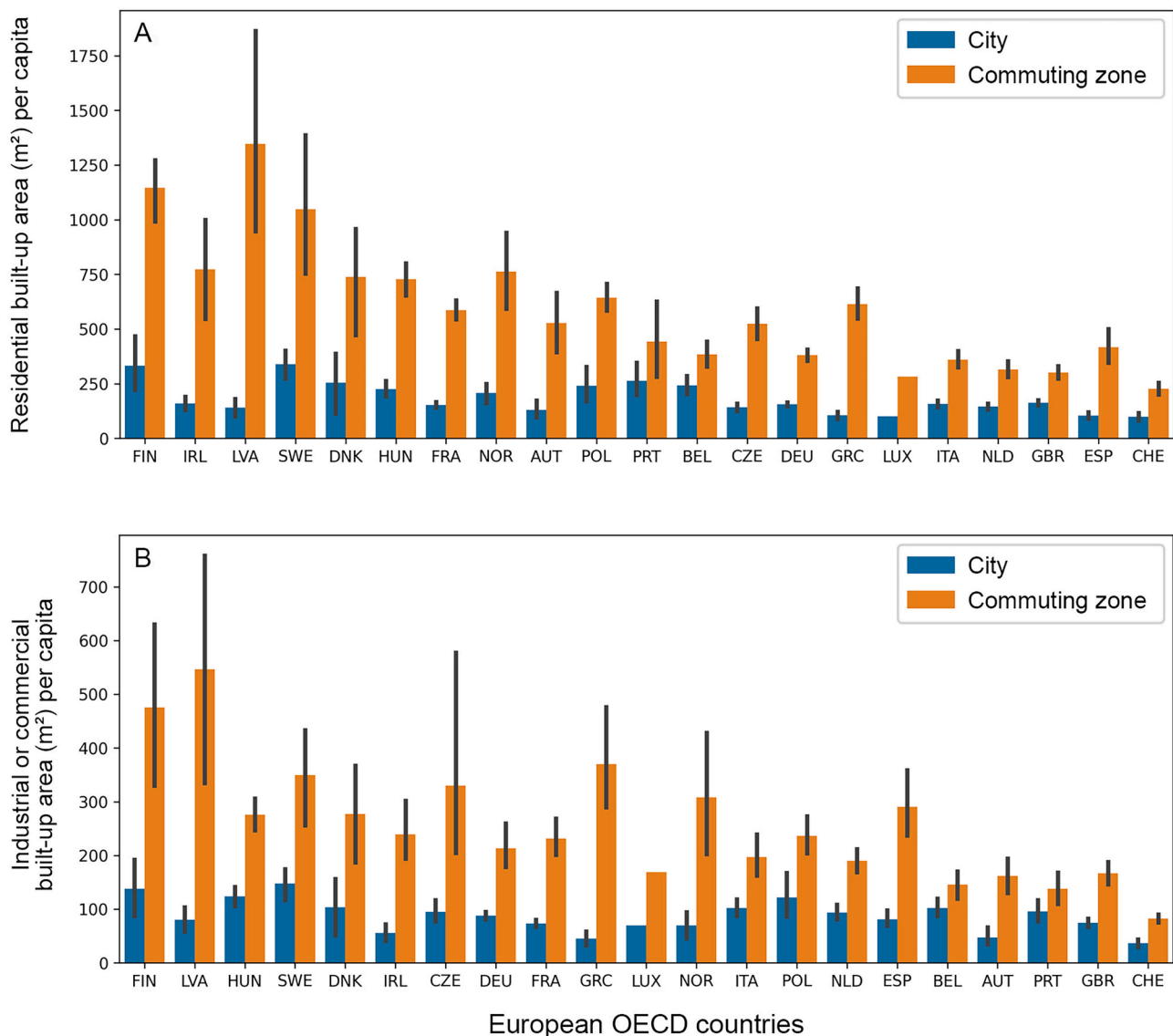


Fig. 9. Built-up area per capita estimates disaggregated by the internal metropolitan structure of cities and commuting zones.

relative differences in built-up area per capita between cities and commuting zones are not merely a matter of extremes among countries; intra-metropolitan differences of similar magnitudes are namely observed for countries of various levels of built-up area per capita, see for instance Norway, Greece, and Spain.

C. Built-up area per capita is typically lower in larger metropolitan areas and show the most variation among small-sized 'metropolitan' areas—In another comparative assessment, built-up area per capita by type of use can be examined for small as well as medium-sized, large, and very large metropolitan areas based on the consistent OECD definition, from Fig. 10. For both observed uses, built-up area per capita tends to be smaller for cities and commuting zones in larger metropolitan areas. This observation is again in line with notions of competition for space by households and firms, which is typically more intensive in larger cities, so leading to built-up areas.

Being used relatively more intensively there. However, also among the largest of metropolitan areas there is substantial variation in residential as well as industrial or commercial built-up area per capita, whereas such variation is highest in the smallest of 'metropolitan' areas.

4. Discussion

The proliferation of remote sensing as a worldwide source of understanding urban areas is clear and driven by a scientifically dynamic and productive field. At the same time, as argued also in a recent multi-decade overview of studies on urban areas, the field may benefit substantially from closer linkages with other disciplines, as cities and other places in the 'urban system' are crucially shaped by social and economic processes (cf., Zhu et al., 2019). In this vein, the present study has importantly introduced an established internationally consistent people-based 'economic' definition of metropolitan areas to the literature on the remote sensing of urban areas. The merits and implications of this are plural and will be discussed next, followed by brief notes on transferability and internationalization, in light of how this study's findings potentially affect the interpretation of outcomes in remote sensing studies.

4.1. Implications

This study's main implication is that, in a context of urban land cover or land use classification, the choice of study area can now be better understood as an essential feature of overall study design in many more

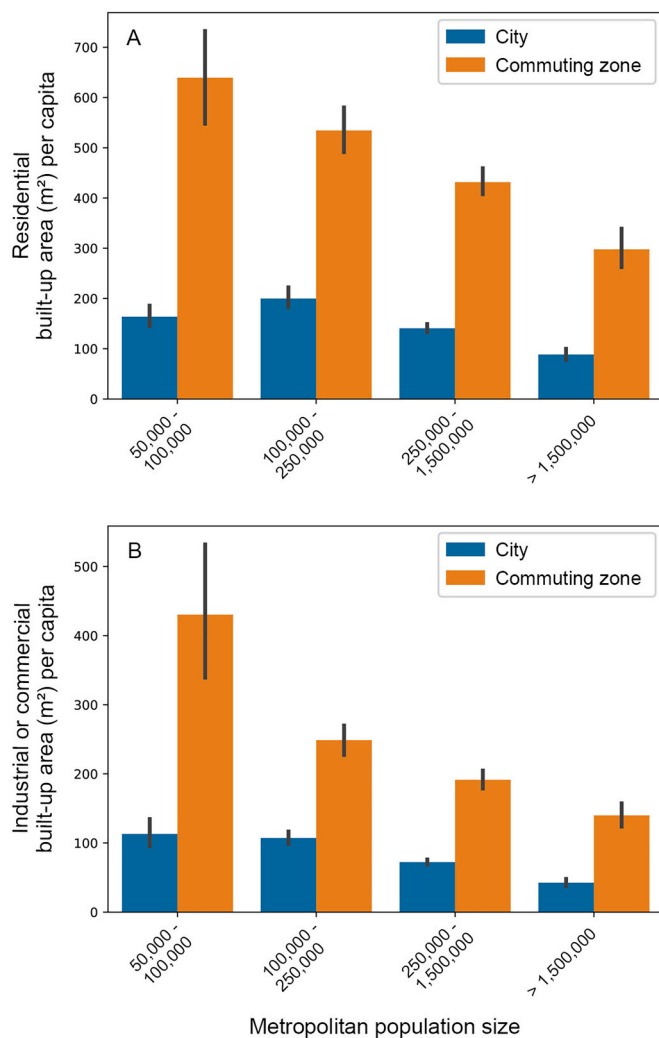


Fig. 10. Built-up area estimates by metropolitan area size-category.

studies of either single or multiple cities or regions. Our findings highlight on a uniquely consistent comparative basis how choice of study area may a priori affect classification outcomes. The accuracy of estimates is namely shown empirically to be potentially related to both the features of metropolitan areas, which is largely a matter of possible study area selection-effects, as well as to the more definitional matter of what type of urban sub-areas are deliberately or arbitrarily enclosed in the study area.

Selection-type of effects may arise from the apparent idiosyncrasy of distinct metropolitan areas, considering the results in Fig. 4 in which the classification scheme is held constant. This suggests that when a study area includes multiple urban areas, the established good practice of adjusting area estimates for mapping errors, see Olofsson et al. (2014), should consider to do so not merely on a study area-wide basis, as in this study would cover Europe, but at the scale of distinct urban areas. This holds in particular when classifications are analyzed for individual urban areas. Moreover, part of the observed metropolitan variation in classification accuracy can be systematically traced to specific metropolitan features. These features include population size and the internal structure in terms of cities and commuting zones, as both categorized for international consistency by the OECD and the European Union, as well as city shape as captured from the ratio of built-up area. Importantly, these accuracy issues may affect not only studies of multiple urban areas but also those of single urban areas. Furthermore, the observations here may hold broader relevance to the interpretation of studies based on

large-scale map products (cf., Brown et al., 2022; Esch et al., 2017; Estoque et al., 2021; Schiavina et al., 2022; Tsendbazar et al., 2021) as also in such instances idiosyncratic or systematic confounders of classification accuracy are plausible, which might warrant further documentation.

Matters of study area definition, on the other hand, may arise as accuracy can vary with internal urban structure. For instance, as would appear to be consistent with the observation that accuracies in rural settlements may be comparatively low (Hoffman-Hall et al., 2019) albeit that, in the instance referred to here, the setting is relatively remote, classifications in cities typically outperform those in commuting zones (Tables A.1 and A.2 in the Appendix). As such, the extent to which cities or commuting zones, or sub-areas of either, are enclosed in a given study's study area, may affect the level of accuracy at which classifications are obtained. Study area definition may have additional implications in particular for the interpretation of area estimates, meaning area surfaces not accuracies. In quantifying the surface of built-up area in a particular use, the spatial definition of a study area may namely substantially influence outcomes, as this study's findings underline in the vein of widely established economic theory and analysis (Duranton and Puga, 2015; Moreno-Monroy et al., 2021). More specifically, the spatial scope of analysis – for instance, regarding whether a small-sized or large-sized metropolitan area is observed, or the extent to which central land or decentral land is included in the study area, or even which country is observed – may influence area estimates (km²) to a considerable extent (cf., Section 4.2). This can be observed in the present study on a consistent basis due to the metropolitan area definition, whereas under study area definitions that are arbitrary, or otherwise idiosyncratic, such differences in area estimates may be amplified or obfuscated.

4.2. Brief notes

This study does not claim that using metropolitan areas is necessarily preferable over other approaches followed in the urban land cover and land use literature, even though economists often prefer this approach (Bosker et al., 2021; Dijkstra et al., 2021; Moreno-Monroy et al., 2021). Urban areas as concentrations of human activity comprise of more than labor markets, see Duranton (2021), as for instance Liu et al. (2015) consider in measuring urban areas from taxi rides. Relevant socio-economic interactions or linkages as well as any different policies by distinct local administrations that shape them may, however, ultimately be enclosed by commuting zones, these being comprehensively wide units. Furthermore, agglomeration-based boundary definitions have merits, aside from potential endogeneity with urban expansion, whereas metropolitan areas, whose measurement typically assumes a minimum population threshold, may overlook small but rapidly growing urban agglomerations, or ignore ghost towns. Administrative boundaries, on their part, may serve to capture relevant scales of regional or local policy processes. Local policies may be noted to shape the efficiency of land use in particular locations, or what locations people or firms sort into. Such processes are overarched by the labor market linkages that functionally integrate sub-areas with different characteristics (Dijkstra et al., 2019; Duranton, 2021). This means that any effects of intra-metropolitan variation in policy are absorbed into our metropolitan-level results, as intended, rather than overlooked or amplified as when an urban study area's boundaries would be defined more narrowly than the internal reshuffling of urban growth and land use. Ultimately, however, the merits of different approaches to defining urban areas depend on a study's objectives.

Furthermore, the evaluation of classification accuracy across different places remains a challenging exercise, in particular in the context of deep learning when the scale of evaluation may not be individual pixels but rather objects or convolutional context (e.g., Chen et al., 2021; Reba and Seto, 2020; Stehman and Foody, 2019; Wickham et al., 2013; Ye et al., 2018). Also, as the present study considers reference classifications based on secondary data, albeit these are partly

generated through intensive manual processes, this may to some extent have influenced the accuracy signal observed in the results, besides possible algorithm-specific matters as discussed in [Reba and Seto \(2020\)](#) in more detail. What may mitigate these issues is that the classification objectives in the present study were based on ambitions that were relatively ‘modest’, as the underlying classifications pertain to ‘mere’ distinct types of land use. Other studies have perhaps more adventurously sought to classify building height (e.g., [Li et al., 2020](#)), local climate zone typologies (e.g., [Yoo et al., 2019](#)), or intra-city zones of various types (e.g., [Du et al., 2021](#)). Although the present study leaves the remote sensing of intra-urban subareas to others, also in light of such classification objectives the issue of ‘what spatial units’ to compare with each other, to further examine cross-metropolitan consistency of comparison, could be compelling.

Regarding an explicitly international setting of country-by-country assessment, the findings in this study have furthermore shown a considerable heterogeneity in remote sensing outcomes for metropolitan areas, both in terms of accuracy and per-capita area estimates. This points to heterogeneity in economic and governance forces as well as possibly confounding aspects of land cover or land use. In this context, it may be noted that the present study has on the one hand exploited the profound heterogeneity of metropolitan areas across Europe, and yet the precise results may only be limitedly transferable to other world regions. This may relate to climatic factors, although these appear to have a relatively limited effect in the present study (see [Fig. A.2](#) in the Appendix) or to different types of urban structure, such as the degree of mixed use of land, which limits the clarity of land use assignment in mixed intra-urban areas in Europe ([Rosier et al., 2022](#)), and is particularly characterizing of metropolitan areas in, for instance, Asia or South America. Also, the present analysis solely observes metropolitan areas in OECD-member countries, which are considered to be developed countries, given the availability of data relevant to the purposes of this study. The consistent definition of metropolitan areas by the OECD and the European Commission, however, has been extended globally such that countries at any stage of development are now covered, see [Moreno-Monroy et al. \(2021\)](#), which would permit for cross-metropolitan analysis, once internationally harmonized reference data are better available, on an even broader basis.² This could practically allow for the improved comparison of classification outcomes, or further assessment of how certain forces may mediate or confound the classification process itself, across the world’s many widely differing national and regional settings. For instance, associations of city characteristics with classification accuracy, or the extent to which amounts of built-up land use are driven by commuting zones, or by the economic or governance features of cities, could even across continents be examined on a more consistent basis using the EU-OECD metropolitan boundaries – for metropolitan areas in any country and, as potentially relevant to many studies, of any size. *Such efforts, in line with the present study’s intentions, might meaningfully aid to further promote generalization in the growing and global field of remote sensing. This, in turn, may help to better inform policy making processes at an increasingly crucial scale of decision-making and sustainability-crossing impacts, see [OECD \(2022\)](#) and [Seto et al. \(2017\)](#), namely the scale of individual metropolitan areas.*

5. Conclusions

This study has examined the usefulness and possible implications of a

consistent measure of metropolitan boundaries for the remote sensing of urban land. In doing so, the OECD-EU definition of metropolitan boundaries was introduced to the literature on remote sensing. More specifically, in a case-study of 687 European metropolitan areas a deep learning model was used to segment several land cover and land use classes in a fusion of radar and optical imagery sourced from EU-ESA Sentinel satellites. Analysis of metropolitan structure in terms of built-up area in either a residential use or an industrial or commercial use shows that *area estimates as well as classification accuracy, in terms of precision-recall profiles and F1-scores, vary widely across distinct metropolitan areas as well as, to some degree, within and between the observed countries. Moreover, while controlling for country-level effects, the accuracy at which built-up areas are classified, by type of use and as measured at the level of individual metropolitan areas, is observed to be positively associated with both population size and general built-up area density.* These observations have potential implications for the interpretation of area estimates and their accuracy in both studies of either single or multiple urban areas. In particular, this study’s findings highlight the typically unobserved role that study area definition and selection may play in affecting the outcomes of remote sensing-based studies of land cover or land use in an urban setting.

CRedit authorship contribution statement

Michiel N. Daams: Conceptualization, Methodology, Investigation, Supervision, Writing – original draft, Writing – review & editing. **Alexandre Banquet:** Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Writing – review & editing. **Paul Delbouze:** Methodology, Data curation. **Paolo Veneri:** Conceptualization, Supervision, Project administration, Funding acquisition, Resources, Writing – review & editing.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: This project was produced within the OECD-EU project on Global State of Cities, financially supported by the European Commission, Directorate-General for Regional and Urban Policy. Michiel N. Daams reports a relationship with OECD that includes consulting.

Data availability

Relevant code and data will be made publicly available in an established repository upon publication.

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² The global OECD-EC metropolitan area boundaries dataset is publicly accessible from: <https://www.oecd.org/cfe/regionaldevelopment/metropolitan-areas.htm>.

Appendix

A. Classification outcomes by internal spatial structure of metropolitan areas

Tables A.1 and A.2 presents confusion matrices for the test set of labelled images when split by the sub-areas of metropolitan areas, namely cities and commuting zones. The observed classification outcomes are for the main model.

Table A.1

Confusion matrix for classification results in cities.

		Reference class						User's acc. (precision)
		Residential	Industrial or commercial	Transport infrastructure	Open space	Agricultural	Water or wetlands	
Predicted class	Residential	66,527,862	6,954,071	9,302,717	4,301,782	4,031,224	97,086	0.73
	Industrial or commercial	3,445,267	29,908,122	4,239,235	3,614,872	2,839,851	173,810	0.68
	Transport	763,559	1,279,709	9,139,220	1,265,815	1,545,612	118,653	0.65
	Open space	2,640,490	2,945,542	2,463,916	211,084,461	26,042,856	2,121,725	0.85
	Agricultural	2,665,376	2,222,715	2,635,060	13,440,243	187,501,642	1,193,214	0.89
	Water or wetlands	58,173	119,372	88,079	906,810	563,124	26,485,135	0.94
	Producer's acc. (recall)	0.87	0.69	0.33	0.90	0.84	0.88	0.84

Table A.2

Confusion matrix for classification results in commuting zones.

		Reference class						User's acc. (precision)
		Residential	Industrial or commercial	Transport infrastructure	Open space	Agricultural	Water or wetlands	
Predicted class	Residential	93,133,340	7,698,581	10,234,185	5,760,955	12,907,023	105,157	0.72
	Industrial or commercial	3,923,185	30,660,153	2,642,231	3,538,194	4,942,333	249,612	0.67
	Transport	754,143	906,506	9,244,146	1,292,653	3,222,034	84,951	0.60
	Open space	4,558,130	4,070,566	5,143,664	664,897,533	59,299,359	5,133,847	0.90
	Agricultural	10,683,800	5,265,858	11,194,968	36,093,706	922,506,551	3,275,871	0.93
	Water or wetlands	89,606	195,757	80,397	1,760,952	1,366,567	46,485,086	0.93
	Producer's acc. (recall)	0.82	0.63	0.24	0.93	0.92	0.84	0.90

The results in Tables A.1 and A.2 indicate that, for both of the land use classes of interest, residential built-up area and industrial or commercial built-up area, classification in cities outperforms classification in commuting zones, in particular in terms of producer's accuracy, whereas non-built-up classes are typically classified with a higher accuracy in commuting zones.

B. Classification outcomes by country and global versus climate zone-specific model

Fig. A.2 shows map classification outcomes based on F1-scores by country and model type. The 'global' model refers to the main model, which is trained on a study area-wide sample of images and reference labels, whereas the alternative model is a set of distinct models that are trained and estimated separately for different climate zones.

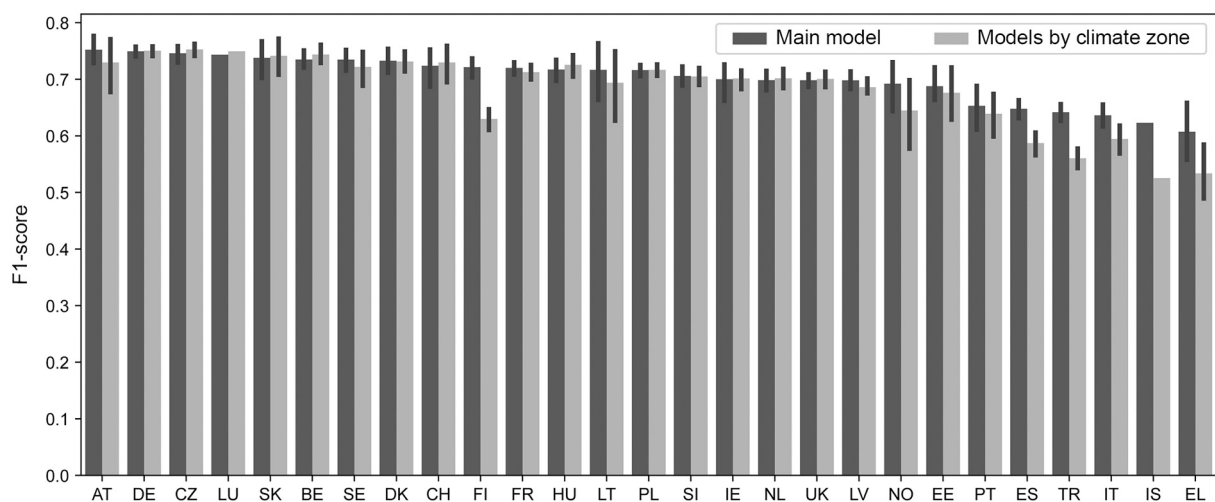


Fig. A.2. Classification accuracy (F1-scores) by country and model type, with distinct outcomes for a pooled 'global' model and across climate zone-specific models.

Climate zones here are defined based on the Köppen-Geiger classification (Beck et al., 2018). It could be noted that sample-splits by climate zone

reduces the sample size observed in model training processes, which may in turn possibly reduce classification accuracy outcomes.

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